

VG4142SxxxX0M1 Wireless Module Hardware Specifications

V1.0



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1. Overview

VG4142SxxxX0M1 series wireless module, based on PANCHIP's PAN3031 high-performance wireless transceiver chip design, is an ultra-long-distance bidirectional wireless transceiver module with built-in power amplifier (PA) and low noise amplifier (LNA). PAN3031 is a low-power long-distance wireless transceiver chip using Chirp-IOT modulation and demodulation technology, supporting half-duplex wireless communication, with an operating frequency band of 400~510MHz/768~1020MHz. The chip has the characteristics of high anti-interference, high sensitivity, low power consumption and ultra-long distance. This module adds a power amplifier (PA) and a low noise amplifier (LNA) on the original basis, so that the maximum transmission power reaches 1W while the receiving sensitivity is further improved, and the overall communication distance and stability are greatly improved compared with ordinary power modules.

This series of modules integrates all RF-related functions and devices. Users do not need to have a deep understanding of RF circuit design, and can use the module to easily develop wireless solutions and wireless IoT devices with stable performance and high reliability.

Main features of the product:

- Chirp-IOT modulation
- Maximum link budget up to 164dB
- Maximum transmit power 30dBm, programmable configuration
- High receiving sensitivity: -134 dBm
- Wide operating voltage range: 4.5~5.5V
- Support bandwidth 125KHz, 250KHz, 500KHz
- Support spreading factor SF: 7~9

Applications:

- Supply chain and logistics
- Smart city
- Smart agriculture
- Industrial remote control
- Street light
- Remote control application
- Building automation
- Security systems

2. Technical Parameters

Parameter	Value range	Remark
Voltage range	4.5~ 5.5V	5V nominal
Frequency range	433MHz, 868MHz, 915MHz	The applicable frequency band is determined by the module model
Crystal frequency	32MHz	Passive crystal oscillator
Maximum output power	30dBm	1000mW
Air data rate	1.04kbps~20.4kbps	Programmable configuration
Modulation	Chirp- IOT	
Receiving sensitivity	- 134dBm	SF=9,BW=125kHz
Channel bandwidth	125kHz, 250kHz, 500kHz	Programmable configuration
TX current	600mA	Transmit power = 30dBm
RX current	30mA	Non-DC-DC Mode
Sleep current	<5uA	
Control interface	SPI	
Antenna impedance	50 Ohm	
Antenna connection	Side stamp hole	
Storage temperature	-55°C~+125°C	
Operating temperature	-40°C~+85°C	Industrial Grade
Dimensions	29.9x20.5mm	

3. Pin location diagram

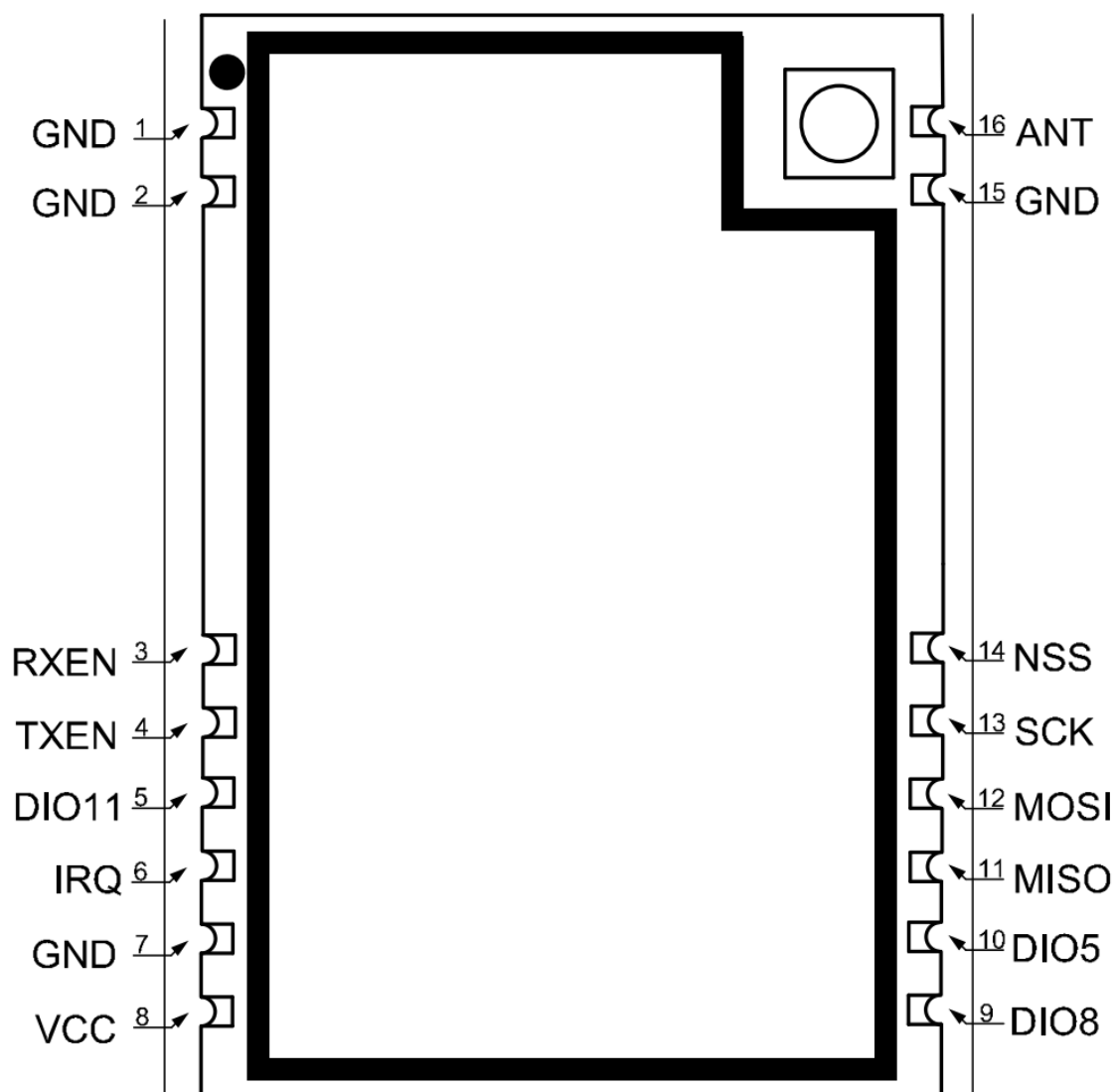


Figure 1-1 Top view

4. Pin Description

No.	Name	Type	Description
1	GND	Power	Ground
2	GND	Power	Ground
3	RXEN	I	LNA control pin, the level control logic when receiving is RXEN=1, TXEN=0 In sleep mode, RXEN=0, TXEN=0
4	TXEN	I	PA control pin, the level control logic when transmitting is TXEN=1, RXEN=0 In sleep mode, RXEN=0, TXEN=0
5	DIO11	I/O	Digital IO, software configurable, directly connected to chip GPIO11
6	IRQ	O	Interrupt signal pin
7	GND	Power	Ground
8	VCC	Power	Power supply
9	DIO8	I/O	Digital IO, software configurable, directly connected to chip GPIO8
10	DIO5	I/O	Digital IO, software configurable, directly connected to chip GPIO5
11	MISO	O	SPI interface MISO data output
12	MOSI	I	SPI interface MOSI data input
13	SCK	I	SPI interface clock input
14	NSS	I	SPI interface chip select input
15	GND	Power	Ground
16	ANT	I/O	RF signal input/output, connect to 50Ω antenna

5. Hardware Design Guidelines and Precautions

5.1. Hardware connection diagram

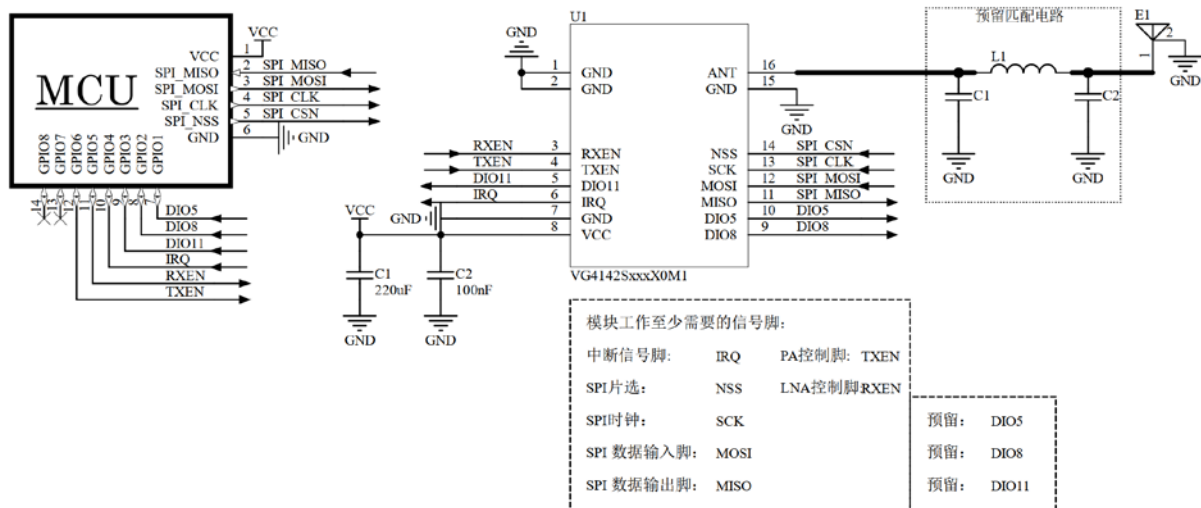


Figure 5-1 Programming development hardware connection

5.2 Power supply design and related considerations

1. Please pay attention to the correct connection of the positive and negative poles of the power supply, and ensure that the power supply voltage is within the recommended power supply voltage range. If it exceeds the maximum allowable power supply range of the module, it will cause permanent damage to the module; the decoupling capacitor of the module power pin should be as close to the module power pin as possible.
2. In the module power supply system, excessive ripple may be coupled to the lines that are susceptible to interference through the wire or ground plane, such as antennas, feeders, clock lines and other sensitive signal lines, which may easily cause the module's RF performance to deteriorate, so we recommend using LDO or linear regulator as the power supply for the wireless module.
3. When selecting LDO or linear regulator chips, it is necessary to pay attention to the heat dissipation of the power supply and the driving ability of the power supply to stabilize the output current; considering the long-term stable operation of the whole machine, it is recommended to reserve more than 50% of the current output margin.
4. It is best to use a separate LDO or linear regulator to power the module; if a DC-DC power chip is used, an LDO or linear regulator can be added later to isolate the module power supply to prevent the noise of the switching power chip from interfering with the working performance of the RF.
5. If the communication line between the MCU and the module uses a 5V level, a 1K-5.1K resistor must be connected in series (not recommended, there is still a risk of damage).
6. Keep the RF module away from high-voltage devices as much as possible, because the electromagnetic waves of high-voltage devices will also have a certain impact on the RF signal.
7. High-frequency digital routing, high-frequency analog routing, and high-current power routing should be kept away from the bottom of the module as much as possible. If it is necessary to pass under the module, the routing must be on another layer of the PCB bottom plate where the module is placed, and ensure that the copper under the module is well grounded.
8. This module is a high-power RF power device. The current consumption is relatively large when the maximum power is transmitted. In order to ensure its high-power transmission driving performance, it is necessary to ensure the stability of the power supply voltage of the module power pin and the current flow capacity of the PCB bottom plate power routing; in addition, the antenna will generate a strong electromagnetic field when the module is

transmitting, to reduce electromagnetic interference to sensitive devices such as the MCU on the baseboard, the RF module and antenna should be kept as far away from sensitive devices on the baseboard as possible.

5.3 Antenna Design Guidance

5.3.1 Stamp hole interface RF design

When the module RF output interface is selected as a stamp hole, use a 50ohm characteristic impedance trace to connect the antenna on the bottom PCB board during design. Considering the attenuation of high-frequency signals, it is necessary to pay attention to the bottom PCB RF trace length should be as short as possible. It is recommended that the longest trace length should not exceed 20mm, and the trace width should be continuous; when turning, try not to use sharp angles or right angles, and it is recommended to use arc lines.

Recommended RF path routing method	
Less recommended RF path routing method	
Bad RF path routing method, not recommended	

To ensure that the impedance of the RF trace on the bottom board is 50 ohms, the following parameters can be adjusted according to the different board thicknesses. The following simulation values are for reference only.

RF routing uses 20mil line width	board thickness 1.0mm, distance between the ground copper and the trace 5.3mil
	board thickness 1.2mm, distance between the ground copper and the trace 5.1mil
	board thickness 1.6mm, distance between the ground copper and the trace 5mil
RF routing uses 25mil line width	board thickness 1.0mm, distance between the ground copper and the trace 6.3mil
	board thickness 1.2mm, distance between the ground copper and the trace 6mil
	board thickness 1.6mm, distance between the ground copper and the trace 5.7mil
RF routing uses 30mil line width	board thickness 1.0mm, distance between the ground copper and the trace 7.6mil
	board thickness 1.2mm, distance between the ground copper and the trace 7.1mil
	board thickness 1.6mm, distance between the ground copper and the trace 6.6mil

5.3.2 Internal Antenna

Built-in antenna refers to the antenna that is welded on the PCB bottom plate and placed inside the product shell, including patch ceramic antenna, spring antenna, etc. When using built-in antenna, the product structure and the installation position of the antenna have a great influence on the RF performance. Under the premise that the product shell structure has enough space, the spring antenna should be placed vertically upward as much as possible; copper cannot be laid around the bottom plate where the antenna is placed, or the circuit board under the antenna can be hollowed out, because metal has a very strong absorption and shielding ability for RF signals, which will seriously affect the communication distance. In addition, the antenna should be placed on the edge of the bottom plate as much as possible.

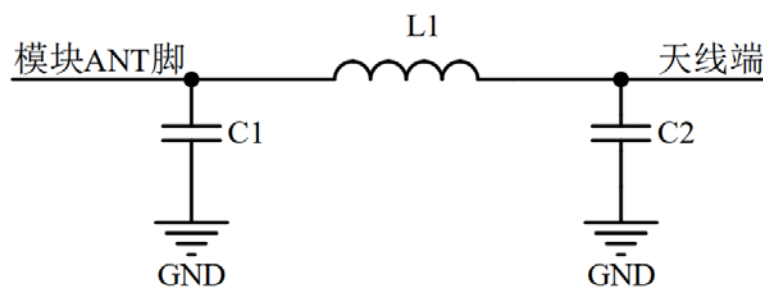
5.3.3 External Antenna

External antenna refers to the antenna installed outside the product housing through IPEX extension cable, SMA and other standard RF interfaces, including rod antenna, suction cup antenna, fiberglass antenna, etc. External antenna is basically a standard product. In order to better choose an antenna suitable for the module, the following should be noted when selecting antenna parameters:

1. The working frequency of the antenna should be consistent with the working frequency of the corresponding module.
2. The input characteristic impedance of the antenna should be 50ohm.
3. The antenna interface size should match that of the module.
4. The antenna standing wave ratio (VSWR) is recommended to be less than 2, and the antenna should have a suitable frequency bandwidth (covering the frequencies used in the actual application of the specific product).

5.3.4 Antenna Matching

The antenna is crucial to the transmission distance of the RF module. In practical applications, in order to facilitate the user's later antenna matching adjustment. It is recommended that the user reserve a simple type matching circuit between the antenna and the module ANT pin output when designing the schematic diagram. If the antenna is already a standard 50Ω, the component L1 is attached with a 0R resistor, and the components C1 and C2 do not need to be soldered. Otherwise, a network analyzer is needed to measure the actual impedance of the antenna and match it to determine the values of C1, L1, and C2. The routing from the module ANT pin to the antenna end should be as short as possible, and the longest routing length is recommended not to exceed 20mm.



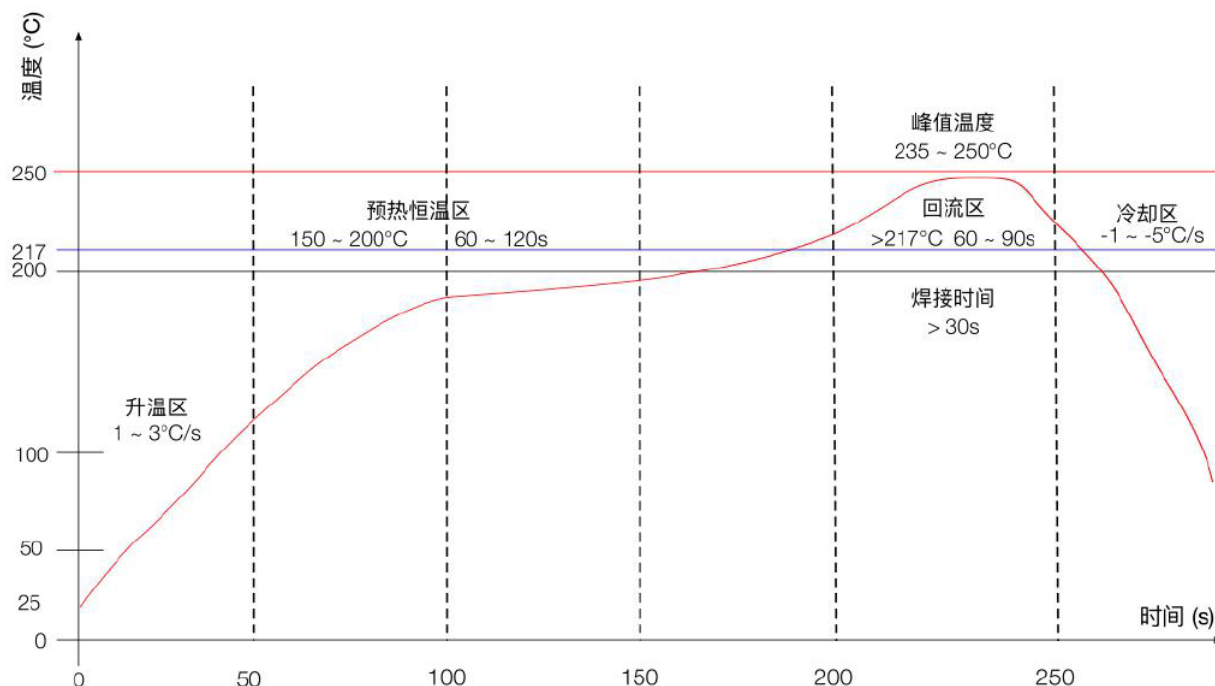
5-2 π -type matching circuit

6. Programming development notes

- 1) Since a PA amplifier is added to the chip, it is recommended that the output power of the front-stage PAN3031 chip be set to a maximum of 15dBm to achieve a PA saturation output power of 30dBm. After the PA reaches saturation power, its power output no longer increases with the increase of the front-stage power, but will increase the power consumption of the module.
- 2) In general, the receiving sensitivity of the RF chip is relatively poor at the integer multiple operating frequency of its crystal oscillator. It is recommended that users pay attention to avoiding the mirror frequency of the module crystal oscillator when selecting the operating frequency, that is, the integer multiple frequency of the crystal oscillator frequency. The crystal oscillator frequency of this module is 32MHz.
- 3) The TXEN and RXEN pins of the module are the logic control pins for controlling the PA&LNA devices inside the module. When using, please pay attention to the control level of the module TXEN and RXEN pins. The control logic is listed as follows:

Mode	TXEN	RXEN
Transmit	1	0
Receive	0	1
Sleep	0	0

7. Reflow Oven Curve



8. Warning of electrostatic damage

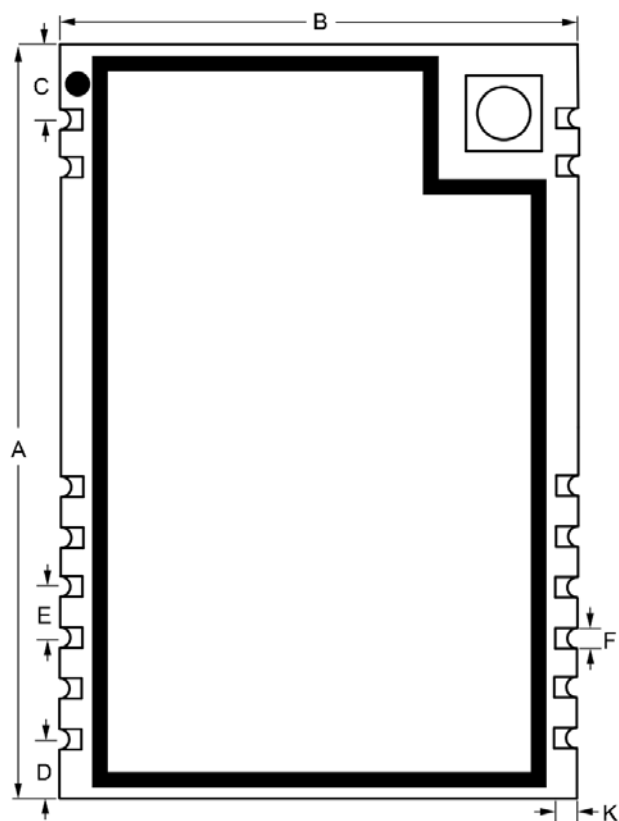
The RF module is a high-voltage electrostatic sensitive device. To prevent static electricity from damaging the module.

1. Strictly follow anti-static measures and do not touch the module with bare hands during the production process.
2. The module should be placed in an area that can prevent static electricity.
3. When designing the product, the anti-static protection circuit at the high-voltage input should be considered.

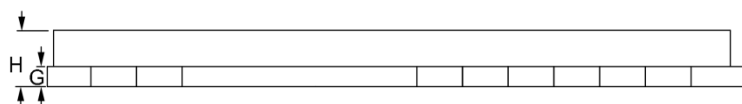


9. Packaging information

Mechanical dimensions (unit: mm)



No.	Dimension (mm)	Tolerance (mm)
A	29.9	± 0.5
B	20.5	± 0.5
C	3.0	± 0.1
D	2.3	± 0.1
E	2.0	± 0.1
F	1.2	± 0.1
K	0.8	± 0.1
G	1.0	± 0.1
H	2.8	± 0.2



10. Version update history

Version	Updated content	Date	Maintainer
V1.0	Initial release	December 3, 2021	Dyming

11. Ordering information

No.	Model	Description
1	VG4142S433X0M1	433MHz band, taping/tray
2	VG4142S868X0M1	868MHz band, taping/tray
3	VG4142S915X0M1	915MHz band, taping/tray

12. Statement

1. Due to product version upgrades or other reasons, the content of this document will be updated from time to time. Unless otherwise agreed, this document is only used as a guide, and the statements, information and suggestions in this article do not constitute any express or implied warranty.

2. The company reserves the final right to interpret and modify all the information provided, and no further notice will be given if there is any change.

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